

PATIENT INPUT: Demographics, Symptoms, Vitals, Medical History

TRI-X Framework Architecture

TRIAGE

Clinical Guidelines

- ACEP Dizziness/Vertigo
- AHA/ASA Stroke
- AAO-HNS BPPV
- Red Flag Detection

TiTrATE

Diagnostic Framework

- Symptom Patterns
- Risk Factor Scoring
- Temporal Analysis
- Comorbidity Assessment

XAI

Explainability Layer

- SHAP Values
- LIME
- NMF Phenotypes
- Counterfactuals
- Rule Extraction

SRGL (Screening-First Risk Governance Logic)

Gate G1: Critical Red Flags → Gate G2: Risk Factors → Gate G3: Uncertainty Quantification

DECISION OUTPUT

Risk Tier (R1-R5) + Urgency Level + Explanation + Care Pathway + Confidence Score

DRAS-5 States

5 Decision-Risk-Action States

ORASR Routing

Safety Routing & Care Pathways

Transparent • Auditable • Safety-First • Clinically-Grounded